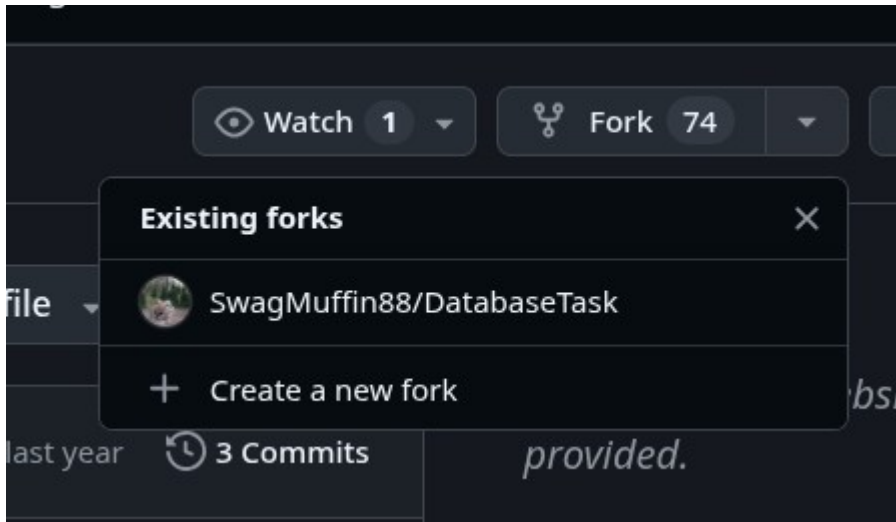


Andmebaasi migratsiooni ülesanne

1. Projekti algne seadistamine

Ülesandega alustamiseks forkisin Githubis õpetaja repo: <https://github.com/IngvarDe/DatabaseTask>



Seejärel kloonisin enda konto all oleva repo oma arvutisse läbi Rider IDE.

Andmebaasi seadistamiseks pidin käivitama MS SQL Docker konteineri:

Search

Only show running containers

	Name	Container ID	Image	Port(s)	CPU (%)	Last	Actions
	mssql_server	008f67e878a1	mssql/server	1433:1433	5.07%	2 da	
	loan-approval-aj	-	-	-	0%	2 m	

Ning samuti pidin veidi ümber muutma projektis olevat andmebaasi seadistust failis appsettings.json, et see vastaks minu süsteemile.

```
c# DatabaseTaskDbContext.cs  global.json  appsettings.json x
1  {
2      "Database": {
3          "Server": "localhost,1433",
4          "Name": "Asd1234"
5      },
6      "Logging": {
```

Edasist seadistust jätkasin Project.cs failis, kus täpsustasin andmebaasi autentimise ja connectionStringi:

```

4
5 > var builder = WebApplication.CreateBuilder(args);
6
7 LoadEnvironmentFile(Path.Combine(builder.Environment.ContentRootPath, "..", ".env"));
8
9 var databaseUsername = Environment.GetEnvironmentVariable("MS_SQL_USERNAME")
10     ?? throw new InvalidOperationException("MS_SQL_USERNAME is not configured.");
11 var databasePassword :string = Environment.GetEnvironmentVariable("MS_SQL_PASSWORD")
12     ?? throw new InvalidOperationException("MS_SQL_PASSWORD is not configured.");
13
14 var connectionString = new SqlConnectionStringBuilder
15 {
16     DataSource = builder.Configuration["Database:Server"],
17     InitialCatalog = builder.Configuration["Database:Name"],
18     UserID = databaseUsername,
19     Password = databasePassword,
20     TrustServerCertificate = true,
21     MultipleActiveResultSets = true
22 }.ConnectionString;
23
24 // Add services to the container.
25 builder.Services.AddControllersWithViews();
26
27 builder.Services.AddDbContext<DatabaseTaskDbContext>(options =>
28     options.UseSqlServer(connectionString));

```

Lisaks on samas failis meetod LoadEnvironmentFile, mis loeb Giti poolt ignoreeritud arenduskeskkonna faili .env-st andmebaasi kasutajanime ja parooli.

Peale seadistust:

- Projekti keeletase on .NET SDK 8.0.127
- Projekt kasutab EF Core 8.0.12
- SQL Server asub aadressil localhost:1433
- Olemas on fail DatabaseTaskDbContext
- DatabaseTask on “startup” projekt
- DatabaseTask.Data on migratsioonide projekt

Ülekontrollimine:

```

~/RiderProjects/DatabaseTask/DatabaseTask master
) dotnet --version
dotnet ef --version
dotnet build DatabaseTask.sln
8.0.127
Entity Framework Core .NET Command-Line Tools
8.0.12
MSBuild version 17.8.49+7806cbf7b for .NET
Determining projects to restore...
All projects are up-to-date for restore.
DatabaseTask.Core -> /home/katri/RiderProjects/DatabaseTask/DatabaseTask.Core
DatabaseTask.Data -> /home/katri/RiderProjects/DatabaseTask/DatabaseTask.Data
DatabaseTask -> /home/katri/RiderProjects/DatabaseTask/DatabaseTask
Build succeeded.

```

2. Code-first migratsioon

Migratsiooni teostamise jaoks oli vaja teha muudatusi olemiklassides nii, et oleks selgelt defineeritud andmebaasi *schemad*. Peale tehtud muudatusi tekkisid olemite vahelised seosed:

- Üks töötaja – mitu last
- Iga laps kuulub ühele töötajale välja EmployeeId kaudu.

Migratsiooni läbiviimiseks kasutasin käsklust:

```
dotnet ef migrations add AddGenderAndChildren \  
    --project DatabaseTask.Data/DatabaseTask.Data.csproj \  
    --startup-project DatabaseTask/DatabaseTask.csproj \  
--context DatabaseTaskDbContext
```

“--project” hoiab migratsioone data-access projektis. “—startup-project” annab migratsioonile vajalikud seadistused ja sõltuvusesisendid (*dependency injection*). “—context” määrab andmebaasikonteksti. Käskluse tulemusena said loodud failid:

- DatabaseTask.Data/Migrations/20260613213420_AddGenderAndChildren.cs
- DatabaseTask.Data/Migrations/20260613213420_AddGenderAndChildren.Designer.cs
- DatabaseTask.Data/Migrations/DatabaseTaskDbContextModelSnapshot.cs

Mida migratsioon teeb (Up() meetod):

- Lisab Employee tabelisse Gender tulba
- Loob tabeli Children
- Lisab Employee.Id võõrvõtmena tabelisse Children
- Loob indeksi Children.EmployeeId jaoks

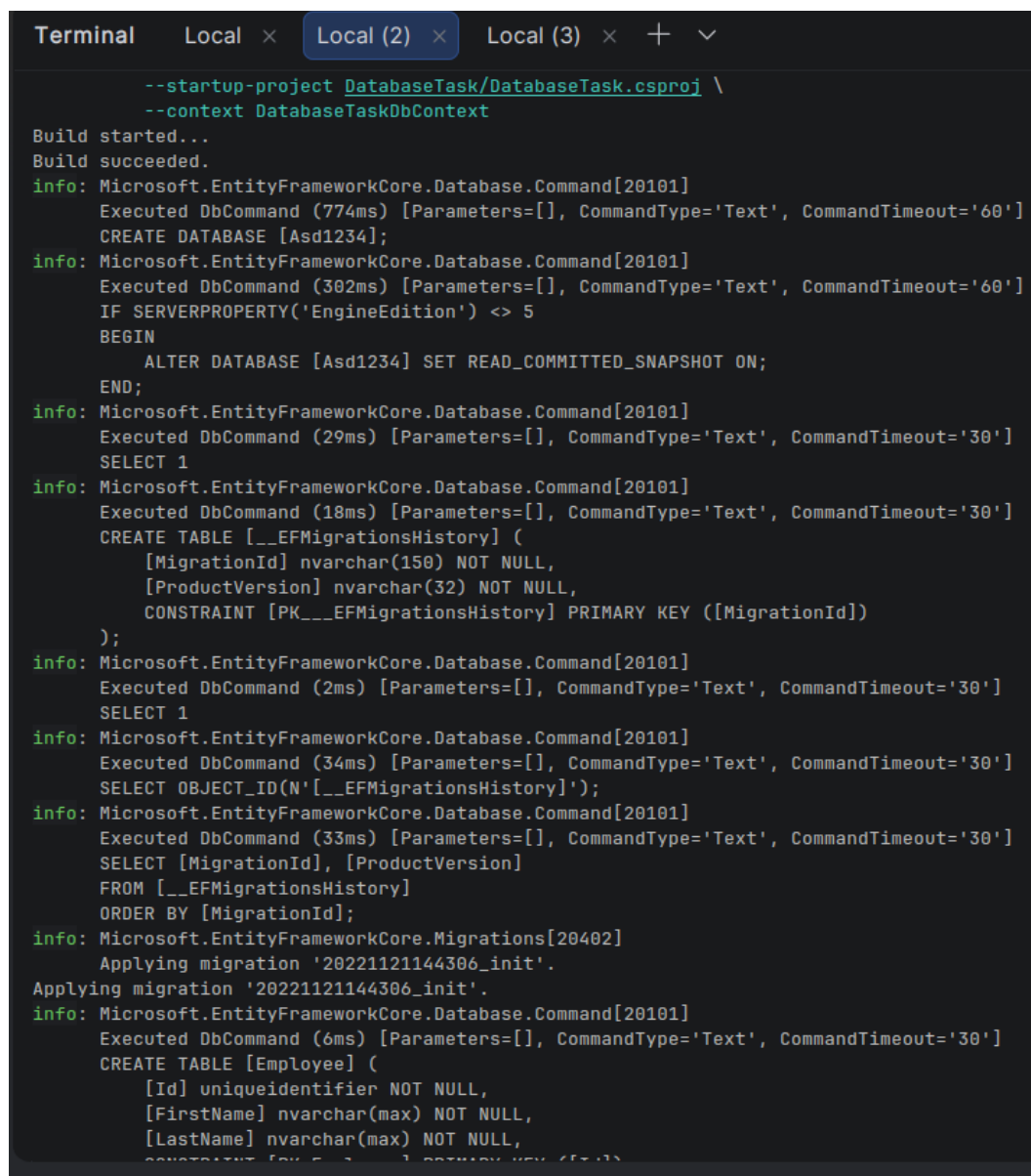
Nüüd on võimalik läbi dotneti lasta genereerida SQL skript, mis näitab, millised muudatused andmebaasis toimusid ja mis on lõpptulemus. Kasutatud käsklus:

```
dotnet ef migrations script \  
    --project DatabaseTask.Data/DatabaseTask.Data.csproj \  
    --startup-project DatabaseTask/DatabaseTask.csproj \  
    --context DatabaseTaskDbContext \  
--output migration.sql
```

Käsklus lõi ülesande juurkausta uue faili migration.sql. Selleks, et andmebaasis ka muudatusi näha, tuli rakendada käsklus:

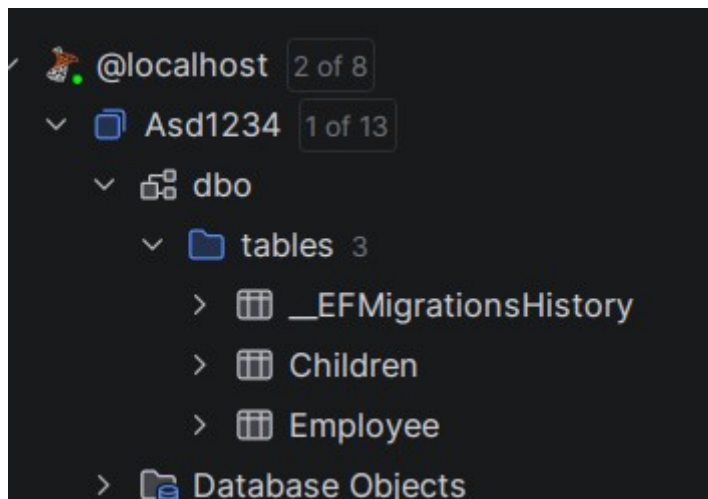
```
dotnet ef database update \  
    --project DatabaseTask.Data/DatabaseTask.Data.csproj \  
    --startup-project DatabaseTask/DatabaseTask.csproj \  
    --context DatabaseTaskDbContext
```

Osaline tulemus konsoolis:

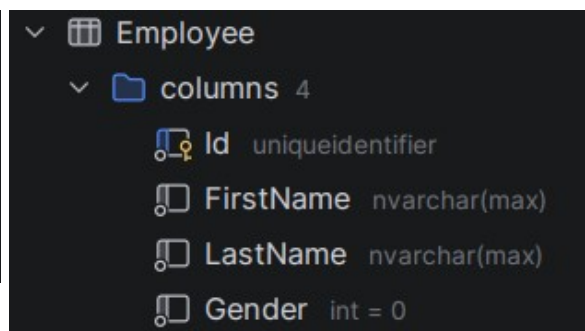
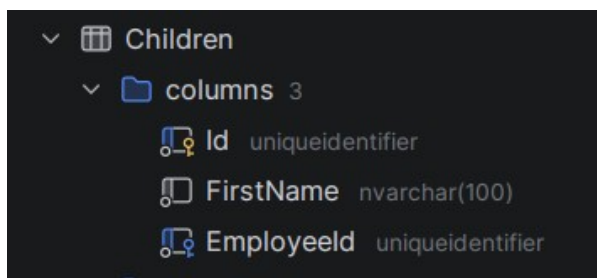


```
Terminal  Local x Local (2) x Local (3) x + v  
--startup-project DatabaseTask.Data/DatabaseTask.Data.csproj \  
--context DatabaseTaskDbContext  
Build started...  
Build succeeded.  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (774ms) [Parameters=[], CommandType='Text', CommandTimeout='60']  
      CREATE DATABASE [Asd1234];  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (302ms) [Parameters=[], CommandType='Text', CommandTimeout='60']  
      IF SERVERPROPERTY('EngineEdition') <> 5  
      BEGIN  
          ALTER DATABASE [Asd1234] SET READ_COMMITTED_SNAPSHOT ON;  
      END;  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (29ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      SELECT 1  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (18ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      CREATE TABLE [__EFMigrationsHistory] (  
          [MigrationId] nvarchar(150) NOT NULL,  
          [ProductVersion] nvarchar(32) NOT NULL,  
          CONSTRAINT [PK__EFMigrationsHistory] PRIMARY KEY ([MigrationId])  
      );  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (2ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      SELECT 1  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (34ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      SELECT OBJECT_ID(N'[__EFMigrationsHistory]');  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (33ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      SELECT [MigrationId], [ProductVersion]  
      FROM [__EFMigrationsHistory]  
      ORDER BY [MigrationId];  
info: Microsoft.EntityFrameworkCore.Migrations[20402]  
      Applying migration '20221121144306_init'.  
Applying migration '20221121144306_init'.  
info: Microsoft.EntityFrameworkCore.Database.Command[20101]  
      Executed DbCommand (6ms) [Parameters=[], CommandType='Text', CommandTimeout='30']  
      CREATE TABLE [Employee] (  
          [Id] uniqueidentifier NOT NULL,  
          [FirstName] nvarchar(max) NOT NULL,  
          [LastName] nvarchar(max) NOT NULL,  
          CONSTRAINT [PK_Employee] PRIMARY KEY ([Id])
```

Projekti andmebaasi struktuuri on nüüd võimalik näha:



Olemitabelite tulbad:



3. Database-first migratsioon

Selle jaoks tegin eraldi git oksa, et säiliks ülesande eelmise osa jaoks tehtud kood. Seejärel lõin juurkaustas uue projekti DatabaseFirst:

```
~/RiderProjects/DatabaseTask/DatabaseTask database-first*
) dotnet new classlib \
  --name DatabaseTask.DatabaseFirst \
  --framework net8.0

dotnet sln DatabaseTask.sln add \
  DatabaseTask.DatabaseFirst/DatabaseTask.DatabaseFirst.csproj
The template "Class Library" was created successfully.

Processing post-creation actions...
Restoring /home/katri/RiderProjects/DatabaseTask/DatabaseTask/DatabaseTask.DatabaseFirst/DatabaseTask.DatabaseFirst.csproj:
Determining projects to restore...
Restored /home/katri/RiderProjects/DatabaseTask/DatabaseTask.DatabaseFirst/DatabaseTask.DatabaseFirst.csproj (in 731 ms).
Restore succeeded.

Project 'DatabaseTask.DatabaseFirst/DatabaseTask.DatabaseFirst.csproj' added to the solution.
```

Veendusin, et on olemas selle etapi jaoks vajalikud paketid (märgitud sinistes kastikestes):

EF	Microsoft.EntityFrameworkCore • 8.0.12	10.0.9
EF	Microsoft.EntityFrameworkCore.Design • 8.0.12	10.0.9
EF	Microsoft.EntityFrameworkCore.SqlServer • 8.0.12	10.0.9
EF	Microsoft.EntityFrameworkCore.Tools • 8.0.12	10.0.9
Implicitly Installed Packages in Solution: 72		

Ja restorisin EF tööriista käsklustega `dotnet tool restore / dotnet ef --version`.

Kuna olin loonud uue oksa ja C# projekti, siis sain lihtsalt luua uued C# klassid *scaffoldimise* abil ilma, et oleksin pidanud hakkama varem tehtud tööd kustutama. Terminali kaudu panin paika seose andmebaasiga koos .env failis olevate kasutajatunnustega ja defineerisin, kuhu peaks andmebaasist loetavad olemid projekti struktuuris minema.

```
set -a
```

```
source .env
```

```
set +a
```

```
CONNECTION_STRING="Server=localhost,1433;Database=Asd1234;User
```

```
Id=${MS_SQL_USERNAME};Password=$
```

```
{MS_SQL_PASSWORD};TrustServerCertificate=True"
```

```
dotnet ef dbcontext scaffold "$CONNECTION_STRING" \
```

```
Microsoft.EntityFrameworkCore.SqlServer \
```

```
--project DatabaseTask.DatabaseFirst/DatabaseTask.DatabaseFirst.csproj \
```

```

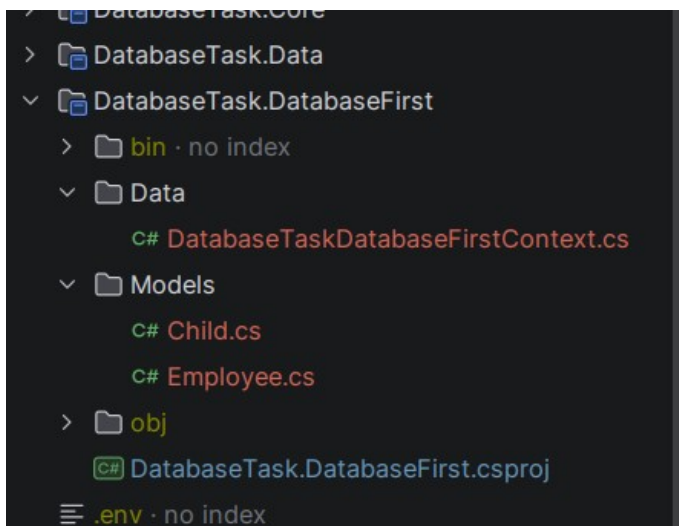
--context DatabaseTaskDatabaseFirstContext \
--context-dir Data \
--output-dir Models \
--no-onconfiguring \
--data-annotations

```

Käsu tulemusel loodi andmebaasi põhjal C# failid:

- DatabaseTask.DatabaseFirst/Data/DatabaseTaskDatabaseFirstContext.cs
- DatabaseTask.DatabaseFirst/Models/Employee.cs
- DatabaseTask.DatabaseFirst/Models/Child.cs

Failid loodi eelmise ülesande etapi jooksul loodud andmebaasi olemite põhjal. Samad failid projekti failipuu:



Võrdlus database-first ja code-first Employee olemiklasside vahel:

